Md. Minhazul Haq

Graduate Research Assistant, Universiti Malaya Portfolio Google Scholar GitHub
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Research Interests

Embedded AI, Smart Transportation System, Lightweight Deep Learning, Computer Vision, UAV Object Detection

Publications

1. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi, Hafiz Muhammad Fahad Noman, and Mohamad Haniff Junos. *HAT-YOLO: Hybrid Attention and Transformer-Integrated Model for UAV Object Detection on Embedded Devices.* KSII Transactions on Internet and Information Systems, 20(2):920–941, 2026.
2. Yang Z, Khairuddin AS, Huang CJ, Ru WW, Noman HM, Putri TW, Xu X, Haq MM. *A Review of Short-Term, Long-Term, and Memory Consolidation Mechanisms in the Hippocampus and Cerebral Cortex.* IEEE Access, 2025.
3. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. *EdgeLite-YOLOv8n: An Efficient YOLOv8n Model for Real-Time Vehicle Detection in UAV Imagery on Embedded Platforms.* Abstract accepted for presentation at the 2nd International Transdisciplinary Research Conference 2026, Universiti Malaya.
4. Md. Minhazul Haq, Anis Salwa Mohd Khairuddin, Effariza Hanafi. *Improved YOLOv8n Model for Automated Wood Texture Defect Detection Based on Computer Vision.* In preparation.
5. Hasan, M. M., K, S., A, M., & Haq, M. M. *Development of Prototype Dual-Axis Solar Tracking Device for Optimal Power Generation at Residential Area in Bangladesh.* Journal of Energy and Safety Technology, 7(1):25–33, 2024.

Poster Presentation: Md. Omar Faruk, Md. Minhazul Haq, Anis Salwa B. M. Khairuddin, Effariza Hanafi, Karen L. Pua, and W. M. Tilakaratne. *Real-Time Oral Cancer Detection Using Lightweight YOLOv8n-Based Deep Learning and Computer Vision: A Pilot Study.* UMHEALTH Academic Month & Research Carnival 2025, Universiti Malaya.

Research Experience

Graduate Research Assistant Nov. 2023 – Feb. 2026
Centre of Intelligent Systems for Emerging Technology (CISET), Universiti Malaya

- Developed lightweight YOLO-based models for UAV object detection on embedded and edge devices.
- Developed and optimized attention-enhanced lightweight computer vision architectures for real-time vehicle and pedestrian detection on a low-cost, resource-constrained Jetson Nano device.
- Conducted benchmarking and comparative evaluation of SOTA object detection models using multiple benchmark datasets.
- Conducted comprehensive literature review on AI-driven smart transportation systems, focusing on lightweight computer vision models for real-time deployment on edge devices.

Education

Master of Engineering Science (Research) Nov. 2023 – Present
Department of Electrical Engineering, Universiti Malaya, Malaysia
Field of Research: Artificial Intelligence
Thesis: *Optimization of Lightweight YOLO Model for Automated Object Detection on Embedded Device*

B.Sc. in Electrical and Electronics Engineering Final Year GPA: 3.55/4.00 | CGPA: 3.11/4.00
Jashore University of Science and Technology (JUST), Bangladesh Graduated: Apr. 2022
Thesis: *Design of Solar Tracking System for Optimal Power Generation at Residential Area in Bangladesh*

Teaching Experience

Graduate Teaching Assistant / Lab Demonstrator Oct. 2025 – Feb. 2026
LAB 2 (KIE1002), Department of Electrical Engineering, Universiti Malaya

Technical Skills

Programming: Python, C/C++, MATLAB, HTML
AI/ML: YOLO, RF-DETR, CNN, RNN, Computer Vision
Tools: OpenCV, PyTorch, Docker, Embedded Systems (e.g., Jetson Nano)

Awards

- **Best Presenter (Postgraduate Oral Presentation) – Winner**, 4th International Orthopaedic Postgraduate Symposium (IOPS4), Universiti Malaya, 29 Oct. 2025.
- **Best Project Award** – Voice-Controlled Humanoid Robot, Science Fair 2019, Jashore University of Science and Technology, Bangladesh.
- **Best Project Award** – Solar Scooter, Student to Startup 2.0 Campus Hunt, JUST, Bangladesh, Oct. 2019.

Standardized Test Scores

IELTS – April 8, 2025: Overall 7.0 (Speaking 7.5, Listening 7.5, Reading 7.0, Writing 6.0)